Oscillations - Approved Segments

30 approved segments, in reading order

SMALL OSCILLATIONS (Chapter 6 Goldstein)

1D CASE: $\ddot{q} = f(q)$ EQUILIBRIUM at $q = q_0$ $\Leftrightarrow$ $0 = f(q_0)$

Now we want to know what happens if we move a little bit away from q_0

$q = q_0 + \eta$ η "SMALL"

$(q_0 + \eta)^{\cdot\cdot} = \ddot{\eta}$ $f(q_0+\eta) = \overset{0}{\cancel{f(q_0)}} + \eta f'(q_0) + \cdots$

$\ddot{\eta} \simeq f'(q_0)\,\eta$ SMALL DEVIATION (EQUATION OF MOTION IS LINEAR IN SMALL DEVIATIONS)

OSCILLATIONS $\Leftrightarrow$ $f'(q_0) < 0$ $\omega = \sqrt{-f'(q_0)}$ $\eta = \eta_0 \cos(\omega t + \varphi_0)$

INSTABILITY $\Leftrightarrow$ $f'(q_0) > 0$ $\lambda = \sqrt{f'(q_0)}$ $\eta = Ae^{\lambda t} + Be^{-\lambda t}$

More than one degree of freedom (Simplify by assuming $V = V(q_1 \cdots q_n, t)$)

$L = T - V$

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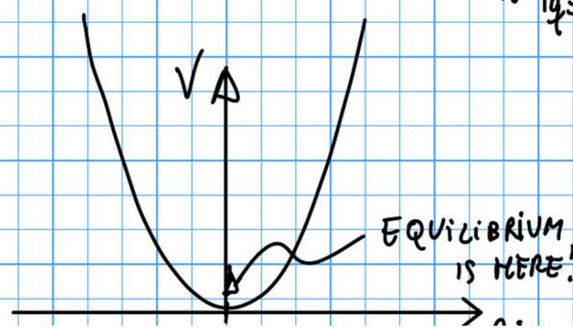
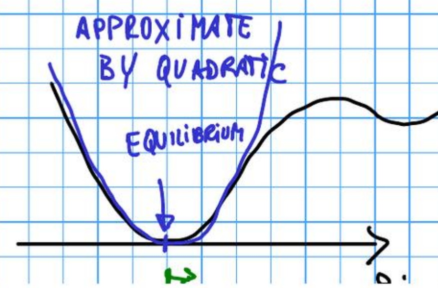
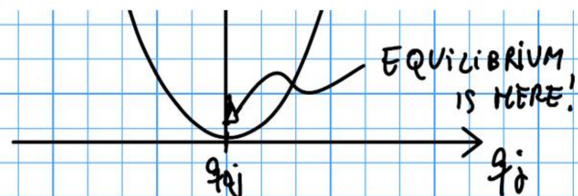
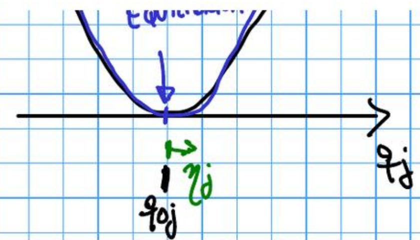
Equilibrium at $(q_1 \dots q_n) = (q_{01} \dots q_{0n})$ $\Longleftrightarrow$ GENERALIZED FORCES $Q_j = -\dfrac{\partial V}{\partial q_j}\Big|_{\vec{q} = \vec{q}_0} = 0$









TAYLOR EXPAND V
TO 2nd ORDER IN η

$q_j = q_{0j} + \eta$

$$V(q_1 \cdots q_n) = \underbrace{V(q_{01} \cdots q_{0n})}_{\substack{\text{CONSTANT} \\ \Rightarrow \text{ NO WORRIES}}} + \underbrace{\sum_{j=1}^{n} \overset{-Q_j = 0}{\cancel{\frac{\partial V}{\partial q_j}}}\Bigg|_{\vec{q}_0} \eta_j}_{0} + \frac{1}{2} \sum_{i,j=1}^{n} \underbrace{\frac{\partial^2 V}{\partial q_i \partial q_j}\Bigg|_{\vec{q}_0}}_{\equiv V_{ij}} \eta_i \eta_j + \cancel{\cdots}$$

$n \times n$ MATRIX
$V_{ij} = V_{ji}$

$q_j = q_{0j} + \eta$

CONSTANT $\Rightarrow$ NO WORRIES

0

$\equiv V_{ij}$

$n \times n$ MATRIX

$V_{ij} = V_{ji}$

$$T = \frac{1}{2} \sum_{ij=1}^{n} \underbrace{\left. \frac{\partial^2 T}{\partial \dot{q}_i \partial \dot{q}_j} \right|_{\substack{\dot{\vec{q}}=0 \\ \vec{q}=\vec{q}_0}}}_{T_{ij}} \dot{\eta}_i \dot{\eta}_j$$

$\dot{\vec{q}} = 0$, $\vec{q} = \vec{q}_0$

$T_{ij} = T_{ji}$

$$\boxed{\begin{aligned} T &= \sum_{p=1}^{N} \frac{m_p}{2} \left(\dot{\vec{r}}_p \right)^2 \\ \dot{\vec{r}}_p &= \sum_{j=1}^{n} \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \cancel{\frac{\partial \vec{r}_p}{\partial t}} \end{aligned}}$$

ASSUME

$$\boxed{L = \frac{1}{2}\sum_{i,j=1}^{n}\left(T_{ij}\,\dot\eta_i\dot\eta_j - V_{ij}\,\eta_i\eta_j\right)}$$

$$\frac{\partial}{\partial\eta_2}\left(\frac{1}{2}\left[V_{11}\,\eta_1\eta_1 + V_{12}\,\eta_1\eta_2 + V_{21}\,\eta_2\eta_1 + V_{22}\,\eta_2\eta_2\right]\right) = \frac{1}{2}\left[\left(V_{12}+V_{21}\right)\eta_1 + 2V_{22}\,\eta_2\right] \overset{V_{12}=V_{21}}{=} V_{21}\,\eta_1 + V_{22}\,\eta_2$$

($\frac{1}{2}$ PREFACTOR CANCELS WITH FACTOR OF 2 IN THE DERIVATIVE)

SIMPLE CASE $T=\frac{m}{2}\dot\eta^2$ $V=\frac{k}{2}\eta^2$

$L=\frac{m}{2}\dot\eta^2-\frac{k}{2}\eta^2$ $\frac{\partial L}{\partial\eta}=-k\eta$ $\frac{\partial L}{\partial\dot\eta}=m\dot\eta$

$0=\frac{\partial L}{\partial\eta}-\frac{d}{dt}\left(\frac{\partial L}{\partial\dot\eta}\right)=-k\eta-m\ddot\eta \Rightarrow \boxed{m\ddot\eta=-k\eta}$

In general: $\frac{\partial L}{\partial\eta_i} = -\sum_{j=1}^{n} V_{ij}\,\eta_j$ $\qquad p_i=\frac{\partial L}{\partial\dot\eta_i}=\sum_{j=1}^{n} T_{ij}\,\dot\eta_j$

Lagrange Eqns: $0=\frac{\partial L}{\partial\eta_i}-\dot p_i \Rightarrow \boxed{0=\sum_{j=1}^{n} -V_{ij}\,\eta_j - \sum_{j=1}^{n} T_{ij}\,\ddot\eta_j}$ $\quad i=1,\dots,n$

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Propose: $\eta_j = \eta_{j0} \cos(\omega t + \delta_j) \Rightarrow \ddot{\eta}_j = -\omega^2 \eta_j \qquad i = 1, \dots, n$

$$0 = \sum_{j=1}^{n} \left(V_{ij}\, \eta_j - \omega^2 T_{ij} \eta_j \right) = \sum_{j=1}^{n} \left(V_{ij} - \omega^2 T_{ij} \right) \eta_{j0} \cos(\omega t + \delta_j) \qquad i = 1, \dots, n$$

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BUT ALSO, since $\cos(\omega t+\varphi_j)\neq 0$ for most times:

$$0=\sum_{j=1}^{n}\left(V_{ij}-\omega^2 T_{ij}\right)\eta_{j0} \qquad i=1,\dots,n$$

EIGENVALUE EIGENVECTOR

ω: NORMAL MODE FREQUENCY

η_{j0}: NORMAL MODE

For the η_{j0} to be nontrivial, we need

$$\boxed{\mathrm{Det}\left[V_{ij}-\omega^2 T_{ij}\right]=0}$$

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ASIDE: More details on the Kinetic Energy part:

$$T = \sum_{p=1}^{N} \frac{m_p}{2}\left(\dot{\vec{r}}_p\right)^2 = \sum_{p=1}^{N} \frac{m_p}{2}\left(\sum_{i=1}^{n} \frac{\partial \vec{r}_p}{\partial q_i}\dot{q}_i + \frac{\partial \vec{r}_p}{\partial t}\right)\cdot\left(\sum_{j=1}^{n} \frac{\partial \vec{r}_p}{\partial q_j}\dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}\right)$$

$$\vec{r}_p = \vec{r}_p(q_1 \cdots q_n, t) \Rightarrow \dot{\vec{r}}_p = \sum_{j=1}^{n} \frac{\partial \vec{r}_p}{\partial q_j}\dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}$$

$$T = T_2 + T_1 + T_0 \qquad T_2 = \frac{1}{2}\sum_{ij=1}^{n} m_{ij}\dot{q}_i\dot{q}_j \qquad m_{ij} \equiv \sum_{p=1}^{n} m_p \frac{\partial \vec{r}_p}{\partial q_i}\cdot\frac{\partial \vec{r}_p}{\partial q_j}$$

$$T_1 = \sum_{j=1}^{n} \pi_j \dot{q}_j \qquad \pi_j \equiv \sum_{p=1}^{n} m_p \frac{\partial \vec{r}_p}{\partial q_j}\cdot\frac{\partial \vec{r}_p}{\partial t}$$

$$T_0 = \sum_{p=1}^{N} \frac{m_p}{2}\left(\frac{\partial \vec{r}_p}{\partial t}\right)^2$$

(m_{ij} is a function of $(q_1 \cdots q_n)$ because the derivatives $\frac{\partial \vec{r}_p}{\partial q_k}$ depend on $(q_1 \cdots q_n)$)

Assuming $\frac{\partial \vec{r}_p}{\partial t} = \vec{0}$, and expanding for $q_j = q_{0j} + \eta_j$ with η_j small, gives

$$T = T_2 = \frac{1}{2} \sum_{i,j=1}^{n} \left(m_{ij}\Big|_{\vec{q}=\vec{q}_0} + \sum_k \frac{\partial m_{ij}}{\partial q_k}\Big|_{\vec{q}=\vec{q}_0} \eta_k + \cdots \right) (q_{0i} + \eta_i)^{\cdot} \, (q_{0j} + \eta_j)^{\cdot}$$

But $(q_{0i} + \eta_i)^{\cdot} = \dot{\eta}_i$ AND $\eta_k \dot{\eta}_i \dot{\eta}_j \propto \eta^3$ IS OF HIGHER ORDER IN SMALL DEVIATIONS

THEREFORE

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THEREFORE, UP TO QUADRATIC ORDER IN $\eta, \dot{\eta}$:

$$\boxed{T = \frac{1}{2} \sum_{i,j=1}^{n} T_{ij}\, \dot{\eta}_i\, \dot{\eta}_j \qquad \text{WITH} \qquad T_{ij} \equiv m_{ij} \Big|_{\vec{q}=\vec{q}_0}}$$

$$\mathrm{Det}\left[V_{ij} - \omega_k^2 T_{ij}\right] = 0 \qquad k=1,\dots,n$$

n # of degrees of freedom

$$\left[\mathbb{V} - \omega_k^2 \mathbb{T}\right] \vec{a}_k = \vec{0} \qquad k=1,\dots,n$$

General solution is linear combination:

$$\eta_i = \sum_k C_k a_{ik} \cos(\omega_k t + \delta_k) \qquad \vec{\eta} = \sum_k C_k \vec{a}_k \cos(\omega_k t + \delta_k)$$

Define $(\mathbb{A})_{ij} \equiv a_{ij}$ $\quad \mathbb{A} \equiv (\vec{a}_1\, \vec{a}_2 \cdots \vec{a}_n)$ $\quad$ <u>Property</u>: $\mathbb{1} = \mathbb{A}^T \mathbb{T} \mathbb{A} \Rightarrow \mathbb{A}^{-1} = \mathbb{A}^T \mathbb{T}$

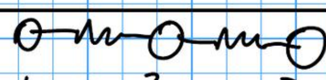
$$\mathbb{A}^T \mathbb{V} \mathbb{A} = \begin{pmatrix} \omega_1^2 & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \omega_n^2 \end{pmatrix}$$

Matrix of eigenvectors/normal modes

$$A^{\dagger} V A = \begin{pmatrix} \omega_1^2 & 0 & \cdots & 0 \\ 0 & \ddots & & 0 \\ 0 & \cdots & 0 & \omega_n^2 \end{pmatrix}$$

Matrix of eigenvectors/normal modes

Normal modes example:



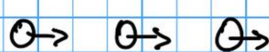
$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

ω_1

$$\vec{a}_2 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

ω_2

$$\vec{a}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$



$\omega_3 = 0$

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = C_1 \vec{a}_1 \cos(\omega_1 t + \delta_1) + C_2 \vec{a}_2 \cos(\omega_2 t + \delta_2) + C_3 \vec{a}_3 \cos(\omega_3 t + \delta_3)$$

$\vec{a}_3 = \begin{pmatrix}1\\1\\1\end{pmatrix}$ $\Theta \rightarrow \quad \Theta \rightarrow \quad \Theta \rightarrow$ $\omega_3 = 0$

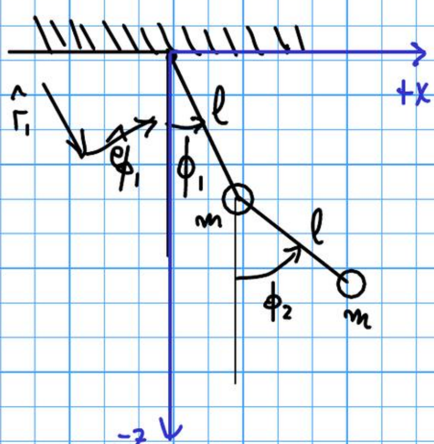
$$\begin{pmatrix}\eta_1\\\eta_2\\\eta_3\end{pmatrix} = C_1 \vec{a}_1 \cos(\omega_1 t + \delta_1) + C_2 \vec{a}_2 \cos(\omega_2 t + \delta_2) + C_3 \vec{a}_3 \cos(\omega_3 t + \delta_3)$$

Normal coordinates $\vec{\zeta}$: $\vec{\eta} = \mathbb{A}\vec{\zeta}$ $\quad \eta_i = \sum_k \zeta_k a_{ik}$ $\quad \zeta_k = C_k \cos(\omega_k t + \delta_k)$ $\quad \vec{\zeta} = \mathbb{A}^{-1}\vec{\eta} = \mathbb{A}^T \mathbb{T} \vec{\eta}$

$$L = \frac{1}{2}\sum_{ij}\left(\dot{\eta}_i T_{ij} \dot{\eta}_j - \eta_i V_{ij} \eta_j\right) = \frac{1}{2}\left[(\dot{\vec{\eta}})^T \mathbb{T} \dot{\vec{\eta}} - \vec{\eta}^T \mathbb{V} \vec{\eta}\right] = \frac{1}{2}\left[\dot{\vec{\zeta}}^T \mathbb{A}^T \mathbb{T} \mathbb{A} \dot{\vec{\zeta}} - \vec{\zeta}^T \mathbb{A}^T \mathbb{V} \mathbb{A} \vec{\zeta}\right]$$

$$L = \frac{1}{2}\left[(\dot{\vec{\zeta}})^T \dot{\vec{\zeta}} - \vec{\zeta}^T \begin{pmatrix}\omega_1^2 & & 0\\ & \ddots & \\ 0 & & \omega_n^2\end{pmatrix} \vec{\zeta}\right] \Rightarrow \boxed{L = \frac{1}{2}\sum_{k=1}^{n}\left(\dot{\zeta}_k^2 - \omega_k^2 \zeta_k^2\right)}$$

Example: Oscillation of a double pendulum



Transformation equations

$$\vec r_1 = l\left(\sin\phi_1\,\hat x - \cos\phi_1\,\hat z\right) = l\,\hat r_1$$

$$\vec r_2 = l\left(\sin\phi_1\,\hat x - \cos\phi_1\,\hat z\right) + l\left(\sin\phi_2\,\hat x - \cos\phi_2\,\hat z\right) = l\left(\hat r_1 + \hat r_{21}\right)$$

$$\hat r_1 = \sin\phi_1\,\hat x - \cos\phi_1\,\hat z \qquad \hat r_{21} = \sin\phi_2\,\hat x - \cos\phi_2\,\hat z$$

Velocities

$$\dot{\hat r}_1 = \dot\phi_1\,\frac{\partial \hat r_1}{\partial \phi_1} \qquad \frac{\partial \hat r_1}{\partial \phi_1} = \hat e_{\phi_1} = \cos\phi_1\,\hat x + \operatorname{sen}\phi_1\,\hat z$$

$$\dot{\hat r}_{21} = \dot\phi_2\,\frac{\partial \hat r_{21}}{\partial \phi_2} \qquad \frac{\partial \hat r_{21}}{\partial \phi_2} = \hat e_{\phi_2} = \cos\phi_2\,\hat x + \operatorname{sen}\phi_2\,\hat z$$

$$\dot{\vec r}_1 = l\,\dot{\hat r}_1 = l\,\dot\phi_1\left(\cos\phi_1\,\hat x + \operatorname{sen}\phi_1\,\hat z\right)$$

$$\dot{\vec r}_2 = l\left(\dot{\hat r}_1 + \dot{\hat r}_{21}\right) = l\,\dot\phi_1\left(\cos\phi_1\,\hat x + \operatorname{sen}\phi_1\,\hat z\right) + l\,\dot\phi_2\left(\cos\phi_2\,\hat x + \operatorname{sen}\phi_2\,\hat z\right)$$

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$$T=\frac{m}{2}\left[(\dot{\vec{r}}_1)^2+(\dot{\vec{r}}_2)^2\right]=\frac{m}{2}\left[l^2\dot\phi_1^2+l^2\left(\dot\phi_1^2+\dot\phi_2^2+2\dot\phi_1\dot\phi_2\,\hat{e}_{\phi_1}\cdot\hat{e}_{\phi_2}\right)\right]\qquad \hat{e}_{\phi_1}\cdot\hat{e}_{\phi_2}=\cos\phi_1\cos\phi_2+\sin\phi_1\sin\phi_2=\cos(\phi_2-\phi_1)$$

$$T=\frac{ml^2}{2}\left[2\dot\phi_1^2+\dot\phi_2^2+2\dot\phi_1\dot\phi_2\cos(\phi_2-\phi_1)\right]\qquad V=mgz_1+mgz_2=-mgl\left(2\cos\phi_1+\cos\phi_2\right)$$

$$L=T-V=\frac{ml^2}{2}\left[2\dot\phi_1^2+\dot\phi_2^2+2\dot\phi_1\dot\phi_2\cos(\phi_2-\phi_1)\right]+mgl\left(2\cos\phi_1+\cos\phi_2\right)$$

We know $\phi_1=\phi_2=0$ is stable equilibrium position, but let's check

$$\frac{\partial V}{\partial\phi_1}=mgl\sin\phi_1=0\ \text{ at }\phi_1=0\ \checkmark\qquad \frac{\partial V}{\partial\phi_2}=mgl\sin\phi_2=0\ \text{ at }\phi_2=0\ \checkmark$$

And V is absolute minimum at $\phi_1=\phi_2=0\ \Rightarrow$ Equil is STABLE

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$\frac{\partial V}{\partial \phi_1} = mgl\sin\phi_1 = 0$ at $\phi_1 = 0$ ✓ $\qquad \frac{\partial V}{\partial \phi_2} = mgl\sin\phi_2 = 0$ at $\phi_2 = 0$ ✓

And V is absolute minimum at $\phi_1 = \phi_2 = 0 \Rightarrow$ Equil is STABLE

$\phi_i^0 = 0$

We thus expand T and V to quadratic order in $\eta_i = \phi_i - \phi_i^0 = \phi_i$

$$\dot\phi_1 \dot\phi_2 \cos(\phi_2 - \phi_1) = \dot\eta_1 \dot\eta_2 \cos(\eta_2 - \eta_1) \underset{\text{small } \eta_i}{\approx} \dot\eta_1 \dot\eta_2 \left[1 - \frac{(\eta_2 - \eta_1)^2}{2} + \cdots \right] \approx \dot\eta_1 \dot\eta_2 + \cancel{O(\dot\eta^2 \eta^2)}$$

$\cancel{\sim \eta^4}$

$$\cos\phi_i = \cos\eta_i = 1 - \frac{\eta_i^2}{2} + \cancel{\frac{\eta_i^4}{4!}} - \cancel{\cdots} \cdots \approx 1 - \frac{\eta_i^2}{2}$$

$T = \frac{1}{2}(\dot\eta_1 \cdots \dot\eta_n)\, \mathbb{T} \begin{pmatrix} \dot\eta_1 \\ \vdots \end{pmatrix}$

$$\cos\phi_i = \cos\eta_i = 1 - \frac{\eta_i^2}{2} + \cancel{\frac{\eta_i^4}{4!}} - \cancel{\cdots} \cdots \approx 1 - \frac{\eta_i^2}{2}$$

$$T = \frac{1}{2}(\dot\eta_1 \cdots \dot\eta_n)\, \mathbb{T} \begin{pmatrix} \dot\eta_1 \\ \vdots \\ \dot\eta_n \end{pmatrix}$$

Thus, to quadratic order in η and $\dot\eta$ we get

$$V = -mgl\left[3 - \frac{2\eta_1^2}{2} - \frac{\eta_2^2}{2}\right] \Rightarrow \mathbb{V} = mgl\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

$$T = \frac{ml^2}{2}\left[2\dot\eta_1^2 + \dot\eta_2^2 + 2\dot\eta_1\dot\eta_2\right] \Rightarrow \mathbb{T} = ml^2\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

CHECK

$$(\eta_1\ \eta_2)\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = (\eta_1\ \eta_2)\begin{pmatrix} 2\eta_1 \\ \eta_2 \end{pmatrix} = 2\eta_1^2 + \eta_2^2 \checkmark$$

$$(\dot\eta_1\ \dot\eta_2)\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \dot\eta_1 \\ \dot\eta_2 \end{pmatrix} = (\dot\eta_1\ \dot\eta_2)\begin{pmatrix} 2\dot\eta_1 + \dot\eta_2 \\ \dot\eta_1 + \dot\eta_2 \end{pmatrix} = 2\dot\eta_1^2 + 2\dot\eta_1\dot\eta_2 + \dot\eta_2^2 \checkmark$$

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To simplify the algebra, we write $\quad mgl = ml^2\left(\frac{g}{l}\right) \qquad \omega_0^2 \equiv g/l$

Eigenvalue equation: $\quad 0 = \mathrm{Det}\left[\mathbb{V} - \omega^2\mathbb{T}\right] = \mathrm{Det}\left(ml^2\begin{bmatrix} 2g/l - 2\omega^2 & -\omega^2 \\ -\omega^2 & g/l - \omega^2 \end{bmatrix}\right)$

$$\Leftrightarrow 0 = \left(2\omega_0^2 - 2\omega^2\right)\left(\omega_0^2 - \omega^2\right) - \omega^4 = \cancel{2}\omega^4 - 4\omega^2\omega_0^2 + 2\omega_0^4 - \cancel{\omega^4} = \omega^4 - 4\omega_0^2\omega^2 + 2\omega_0^4$$

$$\Leftrightarrow 0 = \left(\omega^2 - 2\omega_0^2\right)^2 - \underbrace{\left(4\omega_0^4 - 2\omega_0^4\right)}_{2\omega_0^4} \Leftrightarrow \omega^2 = 2\omega_0^2 \pm \sqrt{2}\,\omega_0^2 = \left(2 \pm \sqrt{2}\right)\omega_0^2 \Leftrightarrow \boxed{\omega = \begin{cases} \omega_+ = \sqrt{2+\sqrt{2}}\,\omega_0 \\ \omega_- = \sqrt{2-\sqrt{2}}\,\omega_0 \end{cases}}$$

COMPLETE SQUARES

Eigenfrequencies

Eigenvectors:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\omega_0^2 - 2\omega^2 & -\omega^2 \\ -\omega^2 & \omega_0^2 - \omega^2 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \Leftrightarrow \begin{aligned} 0 &= 2\left(\omega_0^2 - \omega^2\right)a_1 - \omega^2 a_2 \Leftrightarrow a_2 = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right)a_1 \Leftrightarrow \frac{a_2}{a_1} = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) \\ 0 &= -\omega^2 a_1 + \left(\omega_0^2 - \omega^2\right)a_2 \Leftrightarrow a_1 = \left(\frac{\omega_0^2}{\omega^2} - 1\right)a_2 \Leftrightarrow \frac{a_2}{a_1} = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1\right)} \end{aligned}$$

SIDE NOTE: The 2 equations are equivalent $\Leftrightarrow 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1\right)} \Leftrightarrow 1 = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right)\left(\frac{\omega_0^2}{\omega^2} - 1\right)$

$$\Leftrightarrow \omega^4 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) \Leftrightarrow 0 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) - \omega^4 \Leftrightarrow 0 = \mathrm{Det}(\mathbb{V} - \omega^2 \mathbb{T}) \quad \checkmark$$

So we can take either equation and use it, together with normalization, to get $\vec{a}$:

Propose $\vec{a} = N \begin{pmatrix} 1 \\ 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) \end{pmatrix}$ $\quad N$: normalization constant $\quad x \equiv \frac{\omega_0^2}{\omega^2}$

$$1 = [\vec{a}]^T \mathbb{T} [\vec{a}] = N^2 \begin{pmatrix} 1 & 2(x-1) \end{pmatrix} m\ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2(x-1) \end{pmatrix} = N^2 m\ell^2 \begin{pmatrix} 1 & 2(x-1) \end{pmatrix} \begin{pmatrix} 2x \\ 2x-1 \end{pmatrix}$$

$$1 = N^2 m\ell^2 \left[2x + 2(x-1)(2x-1)\right] = 2N^2 m\ell^2 \left[2x^2 - 3x + x + 1\right] = 2N^2 m\ell^2 \left[2x^2 - 2x + 1\right]$$

$$1 = 2N^2 m\ell^2 \left[2(x-1)^2 + 4x - 2 + 1\right] = 2N^2 m\ell^2 \left\{\underbrace{\left[2(x-1)^2 - 1\right]}_{0} + 2x\right\} = 4N^2 m\ell^2 \frac{\omega_0^2}{\omega^2} \Rightarrow \boxed{N_\pm = \frac{1}{\sqrt{4m}} \frac{\omega_\pm}{\ell\omega_0} = \left(\frac{2\pm\sqrt{2}}{4m\ell^2}\right)^{1/2}}$$

$(x-1)^2 = x^2 - 2x + 1 \quad x^2 = (x-1)^2 + 2x - 1$

$$\omega^4\left[2(x-1)^2 - 1\right] = \left[2(x\omega^2 - \omega^2)^2 - \omega^4\right] = \left[2(\omega_0^2 - \omega^2)^2 - \omega^4\right] = 0$$

$2x^2 - 2x + 1 = 2\left[(x-1)^2 + 2x - 1\right] - 2x + 1 = \left[2(x-1)^2 - 1\right] + 2x$

$$\frac{\omega_0^2}{\omega_\pm^2} = \frac{1}{2\pm\sqrt{2}} = \frac{2\mp\sqrt{2}}{4-2} = 1 \mp \frac{1}{\sqrt{2}} \qquad 2\left[\frac{\omega_0^2}{\omega_\pm^2} - 1\right] = \mp\sqrt{2}$$

$$\boxed{\vec{a}_\pm = \left(\frac{2\pm\sqrt{2}}{4m\ell^2}\right)^{1/2} \begin{pmatrix} 1 \\ \mp\sqrt{2} \end{pmatrix}} \Leftrightarrow \boxed{\omega_\pm = (2\pm\sqrt{2})^{1/2} \frac{g}{\ell}}$$

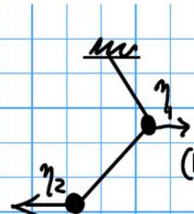
Modal matrix:

$$\boxed{A = \left(\vec{a}_+ \ \vec{a}_-\right) = \left(4ml^2\right)^{-\frac{1}{2}} \begin{pmatrix} a & b \\ c & d \end{pmatrix}}$$

$a \equiv (2+\sqrt{2})^{\frac{1}{2}} = -c/\sqrt{2}$ $\qquad$ $c = -\sqrt{2}\,a = -\sqrt{2}\,(2+\sqrt{2})^{\frac{1}{2}}$

$b \equiv (2-\sqrt{2})^{\frac{1}{2}} = d/\sqrt{2}$ $\qquad$ $d = +\sqrt{2}\,b = +\sqrt{2}\,(2-\sqrt{2})^{\frac{1}{2}}$

Graphical representation:



$\vec{a}_+$

(high frequency ω_+)

$(2-\sqrt{2})^{\frac{1}{2}}(2-\sqrt{2})^{\frac{1}{2}}$

$\vec{a}_-$

(lower frequency ω_-)

$$\left(\begin{array}{c}|\eta_2| = \sqrt{2}|\eta_1| > |\eta_1| \\ \text{IN BOTH} \\ \text{MODES}\end{array}\right)$$

Normal coordinates:

$$\vec{\zeta} = \mathbb{A}^T \mathbb{T} \vec{\eta} = \mathbb{A}^T m l^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

$$2a+c = (2-\sqrt{2})a = \sqrt{2}(\sqrt{2}-1)^{\frac{1}{2}}(\sqrt{2}-1)^{\frac{1}{2}}(2+\sqrt{2})^{\frac{1}{2}} = \sqrt{2}\,2^{\frac{1}{4}}(\sqrt{2}-1)^{\frac{1}{2}}$$

$$a+c = (1-\sqrt{2})a = -(\sqrt{2}-1)^{\frac{1}{2}}(\sqrt{2}-1)^{\frac{1}{2}}(2+\sqrt{2})^{\frac{1}{2}} = -2^{\frac{1}{4}}(\sqrt{2}-1)^{\frac{1}{2}}$$

$$2b+d = (2+\sqrt{2})b = \sqrt{2}(\sqrt{2}+1)^{\frac{1}{2}}(\sqrt{2}+1)^{\frac{1}{2}}(2-\sqrt{2})^{\frac{1}{2}} = \sqrt{2}\,2^{\frac{1}{4}}(\sqrt{2}+1)^{\frac{1}{2}}$$

$$b+d = (1+\sqrt{2})b = (\sqrt{2}+1)^{\frac{1}{2}}(\sqrt{2}+1)^{\frac{1}{2}}(2-\sqrt{2})^{\frac{1}{2}} = 2^{\frac{1}{4}}(\sqrt{2}+1)^{\frac{1}{2}}$$

$$(\sqrt{2}\mp 1)^{\frac{1}{2}}(2\pm\sqrt{2})^{\frac{1}{2}} = 2^{\frac{1}{4}}(\sqrt{2}\mp 1)^{\frac{1}{2}}(\sqrt{2}\pm 1)^{\frac{1}{2}} = 2^{\frac{1}{4}}$$

$$\Big((2a+c)\eta_1 + (a+c)\eta_2\Big)$$

$$\left(\frac{ml^2}{4}\right)^{\frac{1}{2}} = m^{\frac{1}{2}}\frac{l}{2}$$

$$\vec{\zeta} = \mathbb{A}\,\mathbb{T}\,\vec{\eta} = \mathbb{A}\, m\ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

$$b+d = (1+\sqrt{2})\,b = (\sqrt{2}+1)^{\frac{1}{2}}(\sqrt{2}+1)^{2}\,(2-\sqrt{2})^{\frac{1}{2}} = 2^{\frac{1}{4}}(\sqrt{2}+1)^{\frac{1}{2}}$$

$$(\sqrt{2}\mp 1)^{\frac{1}{2}}(2\pm\sqrt{2})^{\frac{1}{2}} = 2^{\frac{1}{4}}(\sqrt{2}\mp 1)^{\frac{1}{2}}(\sqrt{2}\pm 1)^{\frac{1}{2}} = 2^{\frac{1}{4}}$$

$$\vec{\zeta} = (4m\ell^2)^{-\frac{1}{2}} \begin{pmatrix} a & c \\ b & d \end{pmatrix} m\ell^2 \begin{pmatrix} 2\eta_1 + \eta_2 \\ \eta_1 + \eta_2 \end{pmatrix} = \left(\frac{m\ell^2}{4}\right)^{\frac{1}{2}} \begin{pmatrix} (2a+c)\,\eta_1 + (a+c)\,\eta_2 \\ (2b+d)\,\eta_1 + (b+d)\,\eta_2 \end{pmatrix}$$

$$\left(\frac{m\ell^2}{4}\right)^{\frac{1}{2}} = m^{\frac{1}{2}}\frac{\ell}{2}$$

$$\frac{1}{2}\,2^{\frac{1}{4}}(\sqrt{2}\mp 1)^{\frac{1}{2}} = \left(\frac{\sqrt{2}\mp 1}{2\sqrt{2}}\right)^{\frac{1}{2}}$$

$$\boxed{\begin{aligned} \zeta_+(t) &= m^{\frac{1}{2}}\ell\left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right)^{\frac{1}{2}}\left[\sqrt{2}\,\eta_1(t) - \eta_2(t)\right] \\ \zeta_-(t) &= m^{\frac{1}{2}}\ell\left(\frac{\sqrt{2}+1}{2\sqrt{2}}\right)^{\frac{1}{2}}\left[\sqrt{2}\,\eta_1(t) + \eta_2(t)\right] \end{aligned}}$$

But we know also that:

$$\boxed{\begin{aligned} \zeta_+(t) &= f_+ \cos(\omega_+ t + \delta_+) \\ \zeta_-(t) &= f_- \cos(\omega_- t + \delta_-) \end{aligned}}$$

$$\dot{\zeta}_+(t) = m^{\frac{1}{2}}\ell\left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right)^{\frac{1}{2}}\left[\sqrt{2}\,\dot{\eta}_1(t) - \dot{\eta}_2(t)\right]$$

$$\dot{\zeta}_+(t) = -f_+\,\omega_+ \sin(\omega_+ t + \delta_+)$$

$$\boxed{\begin{aligned}\dot{\varphi}_+(t) &= m^{\frac{1}{2}}\ell\left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right)^{\frac{1}{2}}\left[\sqrt{2}\,\dot{\eta}_1(t) - \dot{\eta}_2(t)\right]\\ \dot{\varphi}_-(t) &= m^{\frac{1}{2}}\ell\left(\frac{\sqrt{2}+1}{2\sqrt{2}}\right)^{\frac{1}{2}}\left[\sqrt{2}\,\dot{\eta}_1(t) + \dot{\eta}_2(t)\right]\end{aligned}}$$

$$\boxed{\begin{aligned}\dot{\varphi}_+(t) &= -f_+\,\omega_+\sin(\omega_+ t + \delta_+)\\ \dot{\varphi}_-(t) &= -f_-\,\omega_-\sin(\omega_- t + \delta_-)\end{aligned}}$$

Lagrangian in terms of the normal coordinates:

$$L = \frac{1}{2}\sum_{\alpha}\left(\dot{\varphi}_\alpha^2 - \omega_\alpha^2\varphi_\alpha^2\right) = \frac{1}{2}\left[\left(\dot{\varphi}_+^2 - \omega_+^2\varphi_+^2\right) + \left(\dot{\varphi}_-^2 - \omega_-^2\varphi_-^2\right)\right]$$

$$L = \frac{1}{2}\left[\left(\dot{\varphi}_+^2 - \left[(2+\sqrt{2})\frac{g}{\ell}\right]\varphi_+^2\right) + \dot{\varphi}_-^2 - \left[(2-\sqrt{2})\frac{g}{\ell}\right]\varphi_-^2\right)\right]$$

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Assume initial condition is that at $t=0$: $\quad \phi_1 = \phi_2 = 0 \qquad \dot\phi_1 = -\dot\phi_2 = \nu > 0$

Get $\phi_1(t)$, $\phi_2(t)$ $\qquad$ Then

$$f_+ \cos(\delta_+) = \xi_+(0) = 0 \Rightarrow \delta_+ = \pm\frac{\pi}{2}, \pm\frac{3}{2}\pi, \cdots \qquad m^{\frac{1}{2}} \ell \left(\frac{\sqrt{2}\mp 1}{2\sqrt{2}}\right)^{\frac{1}{2}} (\sqrt{2}\pm 1)\nu = \left(\frac{\sqrt{2}\pm 1}{2\sqrt{2}}\right)^{\frac{1}{2}} m^{\frac{1}{2}} \ell \nu = \sqrt{2\pm\sqrt{2}}\,\frac{m^{\frac{1}{2}}\ell\nu}{2}$$

$$f_- \cos(\delta_-) = \xi_-(0) = 0 \Rightarrow \delta_- = \pm\frac{\pi}{2}, \pm\frac{3}{2}\pi, \cdots$$

Choose $\delta_+ = \delta_- = -\frac{\pi}{2}$

$$-f_+ \omega_+ \sin(\delta_+) = \dot\xi_+(0) = m^{\frac{1}{2}} \ell \left(\frac{\sqrt{2}-1}{2\sqrt{2}}\right)^{\frac{1}{2}} \left[\sqrt{2}\nu - (-\nu)\right] = \sqrt{2+\sqrt{2}}\,\frac{m^{\frac{1}{2}}\ell\nu}{2}$$

$$-f_- \omega_- \sin(\delta_-) = \dot\xi_-(0) = m^{\frac{1}{2}} \ell \left(\frac{\sqrt{2}+1}{2\sqrt{2}}\right)^{\frac{1}{2}} \left[\sqrt{2}\nu + (-\nu)\right] = \sqrt{2-\sqrt{2}}\,\frac{m^{\frac{1}{2}}\ell\nu}{2}$$

$$\Rightarrow f_\pm \sqrt{2\pm\sqrt{2}}\sqrt{\frac{g}{\ell}} = \sqrt{2\pm\sqrt{2}}\,\frac{m^{\frac{1}{2}}\ell\nu}{2} \Rightarrow \boxed{f_\pm = \frac{\nu}{2}\left[\frac{m\ell^3}{g}\right]^{\frac{1}{2}}}$$

$$\cos\left(\omega t - \frac{\pi}{2}\right) = \cos\omega t \,\cancel{\cos\frac{\pi}{2}}^{\,0} + \sin\omega t \,\underset{1}{\sin\frac{\pi}{2}} = \sin\omega t$$

$$\xi_\pm(t) = f_\pm \cos\left(\omega_\pm t - \frac{\pi}{2}\right) = f_\pm \sin(\omega_\pm t)$$

$$a \equiv (2+\sqrt{2})^{\frac{1}{2}} = 2^{\frac{1}{4}}(\sqrt{2}+1)^{\frac{1}{2}} \qquad c \equiv -\sqrt{2}\,a \qquad f_+ a = \frac{\nu}{2}\left[\frac{m\ell^3}{g}\right]^{\frac{1}{2}}(2+\sqrt{2})^{\frac{1}{2}}$$

$$b \equiv (2-\sqrt{2})^{\frac{1}{2}} = 2^{\frac{1}{4}}(\sqrt{2}-1)^{\frac{1}{2}} \qquad d \equiv +\sqrt{2}\,b \qquad f_- b = \frac{\nu}{2}\left[\frac{m\ell^3}{g}\right]^{\frac{1}{2}}(2-\sqrt{2})^{\frac{1}{2}}$$

$$\boxed{\xi_\pm(t) = \frac{\nu}{2}\left[\frac{m\ell^3}{g}\right]^{\frac{1}{2}} \sin\left(\left[(2\pm\sqrt{2})\frac{g}{\ell}\right]^{\frac{1}{2}} t\right)}$$

$$(4m\ell^2)^{-\frac{1}{2}}\frac{\nu}{2}\left(\frac{m\ell^3}{g}\right)^{\frac{1}{2}} = \frac{\nu}{4\sqrt{\frac{g}{\ell}}}$$

In terms of the original coordinates

$$\vec{\eta}(t) = \mathbb{A}\vec{\xi}(t) = (4m\ell^2)^{-\frac{1}{2}}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} \xi_+(t) \\ \xi_-(t) \end{pmatrix} = (4m\ell^2)^{-\frac{1}{2}}\begin{pmatrix} a\,\xi_+(t) + b\,\xi_-(t) \\ c\,\xi_+(t) + d\,\xi_-(t) \end{pmatrix}$$

$$\boxed{\begin{aligned} \phi_1(t) &= \eta_1(t) = \frac{\nu}{4\sqrt{\frac{g}{\ell}}}\left\{(2+\sqrt{2})^{\frac{1}{2}}\sin\left(\left[(2+\sqrt{2})\frac{g}{\ell}\right]^{\frac{1}{2}}t\right) + (2-\sqrt{2})^{\frac{1}{2}}\sin\left(\left[(2-\sqrt{2})\frac{g}{\ell}\right]^{\frac{1}{2}}t\right)\right\} \\ \phi_2(t) &= \eta_2(t) = \frac{\sqrt{2}\,\nu}{4\sqrt{\frac{g}{\ell}}}\left\{-(2+\sqrt{2})^{\frac{1}{2}}\sin\left(\left[(2+\sqrt{2})\frac{g}{\ell}\right]^{\frac{1}{2}}t\right) + (2-\sqrt{2})^{\frac{1}{2}}\sin\left(\left[(2-\sqrt{2})\frac{g}{\ell}\right]^{\frac{1}{2}}t\right)\right\} \end{aligned}}$$

Check initial condition:
At t=0, all sin(...) terms are zero, so we have:

$$\phi_1 = \phi_2 = 0 \quad \checkmark$$

Check initial condition: At t=0, all cos(...) terms are 1, so we

$$\dot{\phi}_1(t) = \dot{\eta}_1(t) = \frac{v}{4}\left\{(2+\sqrt{2})\cos\left(\left[(2+\sqrt{2})\frac{g}{l}\right]^{1/2} t\right) + (2-\sqrt{2})\cos\left(\left[(2-\sqrt{2})\frac{g}{l}\right]^{1/2} t\right)\right\}$$

$$\dot{\phi}_2(t) = \dot{\eta}_2(t) = \frac{\sqrt{2}\,v}{4}\left\{-(2+\sqrt{2})\cos\left(\left[(2+\sqrt{2})\frac{g}{l}\right]^{1/2} t\right) + (2-\sqrt{2})\cos\left(\left[(2-\sqrt{2})\frac{g}{l}\right]^{1/2} t\right)\right\}$$

Check initial condition: At t=0, all cos(...) terms are 1, so we have:

$$\dot{\phi}_1 = \frac{v}{4}\left\{(2+\sqrt{2}) + (2-\sqrt{2})\right\} = v \quad ✓$$

$$\dot{\phi}_2 = \frac{\sqrt{2}\,v}{4}\left\{-(2+\sqrt{2}) + (2-\sqrt{2})\right\} = -v \quad ✓$$

COMPARISON WITH QUANTUM MECHANICS:

QM

$$\begin{array}{ccc} |\psi(0)\rangle & \longrightarrow & \{c_\alpha\} \\ & & \downarrow \\ |\psi(t)\rangle & \longleftarrow & c_\alpha e^{-iE_\alpha t/\hbar} \end{array}$$

Small osc

$$\begin{array}{ccc} \vec{\eta}(0),\ \dot{\vec{\eta}}(0) & \longrightarrow & \vec{\xi}(0),\ \dot{\vec{\xi}}(0) \\ & & \downarrow \\ \vec{\eta}(t),\ \dot{\vec{\eta}}(t) & \longleftarrow & \vec{\xi}(t),\ \dot{\vec{\xi}}(t) \end{array}$$

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$$|\psi(0)\rangle = \sum_\alpha c_\alpha |\varphi_\alpha\rangle \qquad c_\alpha = \langle \varphi_\alpha | \psi(0)\rangle$$

$$\vec{\eta}(t) = \mathbb{A}\,\vec{\xi}(t) \qquad \vec{\xi}(t) = \mathbb{A}^T \mathbb{T}\,\vec{\eta}(t)$$

$$|\psi(t)\rangle = \sum_\alpha c_\alpha e^{-iE_\alpha t/\hbar} |\varphi_\alpha\rangle$$

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